

SwipeHired — Product Requirements Document

Production Specification for AI-Native Recruitment, Career Radar Swiping & 72-Hour Blind Reverse Hiring Marketplace

DOCUMENT VERSION
Version 1.1.0

DOCUMENT STATUS
Production / Engineering Ready

ENGINEERING OWNERSHIP
SwipeHired Core Engineering

TARGET RELEASE SCOPE
Q1 / Q2 2026

1 Document Control & Governance

This document is the authoritative Product Requirements Document (PRD) for **SwipeHired**, superseding Version 1.0.0. It establishes hardened security boundaries, mathematical scoring normalization, privacy data tiers, and testable acceptance criteria across the entire product ecosystem.

2 Executive Summary & Vision

SwipeHired connects engineering talent with high-growth companies through an AI-powered swipe matching deck, Kanban pipeline management, and an anonymous 72-hour blind reverse marketplace. It eliminates application friction, provides candidate transparency, and accelerates hiring velocity.

< 14 Days

TIME-TO-OFFER TARGET

70%

SCREENING TIME REDUCTION

< 5%

TARGET GHOSTING SLA

Zero-PII

BLIND UNTIL REVEAL

Architecture Commitments: PII-protected blind marketplace with zero intentional PII exposure before candidate-authorized bid reveal, multi-attribute algorithmic match scoring, tamper-evident placement agreements, and guaranteed anti-ghosting SLA tracking.

3 Product Strategy & Measurable KPIs

Metric	Definition	Measurement Method	Target	Frequency
Onboarding Velocity	Time from resume upload to completed candidate profile.	Client telemetry timer on parsing workflow.	< 60 Seconds	Continuous telemetry
Match Precision	Progression rate for high-score matches (≥80%).	Database stage progression conversion.	> 85% Precision	Monthly review
SLA Compliance	Applications resolved or progressed within company SLA.	Backend SLA deadline evaluation trigger.	> 95% of Apps	Daily automated scan
Marketplace Acceptance	Ratio of upfront compensation bids accepted vs declined/expired.	<code>talent_bids</code> status aggregation.	> 40% Accept Rate	Weekly telemetry
Agreement Completion	Candidates with tamper-evident e-signatures before radar access.	Candidate gateway authorization checks.	100% Compliance	Per registration

*Benchmarks represent internal product objectives and initial engineering targets rather than guaranteed contractual outcomes.

4 User Personas

Candidate (Alex)

Goals: Fast discovery, salary transparency, blind bias-free evaluation, guaranteed SLA updates.

Pain Points: Lengthy forms, recruiter ghosting, hidden compensation.

Permissions: Edit own profile, sign agreement, swipe jobs, manage applications, list blind profile, accept/counter bids.

Success: Relevant offer in <14 days.

Recruiter (Priya)

Goals: Source verified talent, compare candidates head-to-head, automate interview kits & offer letters.

Pain Points: Inbound spam, disjointed ATS/email tools, candidate drop-offs.

Permissions: Post/manage jobs, browse blind marketplace, place bids, manage Kanban, connect SMTP, generate offers.

Success: 70% screening velocity reduction.

Platform Admin

Goals: Security governance, moderate spam jobs, monitor SLA health, audit placement agreements.

Pain Points: Fraudulent job posts, abuse reports, SLA breaches.

Permissions: Telemetry inspection, user/company status overrides, job moderation, dispute resolution.

Success: 99.9% uptime, zero PII leaks.

5 User Journeys

1

AI Resume OCR

2

Sign Agreement

3

Career Radar Swipe

4

Blind Talent Bidding

5

Kanban ATS & SLA

6

Offer Dispatch

6 Core Product Principles

- Speed & Automation:** Prioritize sub-second responsiveness and automated data extraction over manual entry.
- Privacy by Design:** Enforce zero intentional PII exposure in the blind marketplace until explicit candidate authorization.
- Anti-Ghosting Accountability:** Provide transparent stage tracking and structured constructive feedback for every applicant.
- Security Separation:** UI-level route guards are for user experience only. Database and API authorization define the security boundary.

7 Authentication & Authorization Architecture

Frontend Route Guards + Cloudflare API Authorization + Supabase Row Level Security (RLS) + PostgreSQL Constraints.

- Immutable Roles:** candidate, company, and admin. Role modification blocked server-side.
- Cross-Tenant Isolation:** Recruiters cannot query or mutate data belonging to other companies.
- Agreement Gateway:** Candidates are locked from active swiping and blind marketplace bidding until a valid signed agreement record is verified server-side.

8 Candidate Experience & Onboarding

Candidates progress through a streamlined onboarding flow: Resume Upload → OCR Extraction → Profile Review & Strength Gauge (0–100%) → Placement Agreement Signing → Career Radar Swiping.

9 AI Resume Parsing & Profile Builder

15 Structured Parameters

fullName, headline, email, phone, location, workPreference, yearsOfExperience, skills[], possibleRoles[], education[], experience[], projects[], certifications[], expectedSalary, bio.

OCR & Schema Reliability

Supports PDF/DOCX/TXT/MD up to 25MB. Dual-stage Affinda OCR and LLM schema validation with raw text fallback to prevent UI crashes.

10 Placement Commission Agreement & E-Signature

Candidates execute a digital placement agreement acknowledging a 10% first-year placement commission:

- Tamper-Evident Record:** Document ID SH-AGR-YYYY-[HEX] and SHA-256 integrity hash: SHA256-[HASH1]-[HASH2].
- Signing Audit Trail:** Captures Signer Name, Timestamp, Signature Style, and Client IP Stamp where legally permissible.
- Legal Disclaimer:** Legal enforceability depends on applicable jurisdiction, consent flow, and electronic-signature disclosures. Legal review required before production enforcement.

11 Career Radar & Algorithmic Matching Engine

The matching engine evaluates compatibility using a deterministic multi-attribute scoring model:

RawScore = Base(55) + S_skills(35) + R_role(15) + L_loc(10) + E_exp(8) + C_salary(8) + W_mode(5) [Max Raw: 136]
NormalizedScore = (RawScore / 136) * 100
Rule: Requires ≥1 matching skill OR domain overlap to qualify (Clamped: 60% – 98%). Sub-qualified clamped to ≤40%.

Deterministic Match Score

Authoritative calculation computed server-side from verified profile attributes and job requirements.

Personalization Layer (Recommendations)

Learned Swipe Preferences track candidate actions (apply, skip, save) to re-rank card presentation without altering the objective score.

12 Reverse Blind Hiring Marketplace & 72-Hour Bidding

3-Tier Candidate Privacy Model

- 1. Private Profile:** Full candidate record (name, contact info, resume). Accessible only by the candidate and authorized admin.
- 2. Public Profile:** Intentionally shared professional profile attributes.
- 3. Blind Marketplace Profile:** Anonymous handle (e.g. "Anonymous Full Stack Architect #704"), verified skills, YOE, proof-of-work metrics, salary expectations. Zero contact details.

72-Hour Server SLA Source of Truth: Marketplace bids use server-stamped created_at and expires_at timestamps. Expired bids are rejected server-side regardless of client clock state. Counter-offers preserve full audit history.

13 Recruiter ATS, AI Job Architect & Kanban Pipeline

AI Job Spec Architect

Generates complete job postings from a single-sentence brief, automatically populating department, responsibilities, required skills, and salary bands.

6-Stage Kanban Pipeline

Visual drag-and-drop workflow: Applied → Screening → Shortlisted → Interview → Offer → Hired with fast-action SLA badges.

14 Candidate Comparison Arena

Enables recruiters to evaluate 2 to 4 candidates head-to-head against any target opening with D3/SVG radar overlays and an AI Comparative Verdict analyzing tradeoffs and top picks.

15 Anti-Ghosting SLA & Constructive Feedback

- Company SLA Badges: Public tiers: 🏆 Gold (<24h), 🥈 Silver (<48h), 🥉 Bronze (<72h).
- Live Pulse Steps: Chronological milestone progression visible to candidates.
- Constructive Rejection Plan: Automated gap analysis and 30-day learning roadmap upon application rejection.

16 Multi-Channel Communication Architecture

Platform Transactional Emails

Sent via Cloudflare Email Service / platform worker for signup verification, bid alerts, SLA updates, and password resets.

Recruiter Connected Mailbox & WhatsApp

Recruiter-controlled sender addresses via custom SMTP/Gmail connection testing or direct WhatsApp Web templates. Secrets never exposed to client.

17 AI Interview Kit & Offer Letter Suite

- AI Interview Kit: Generates 5–8 role-tailored technical & behavioral questions with structured evaluation rubrics (Below / Meets / Exceeds).
- AI Offer Letter: Auto-populates candidate details, CTC breakdown, joining dates, and terms for direct dispatch.

18 Notifications & Real-Time Telemetry

In-app real-time notification engine dispatching alerts for matches, bids, stage transitions, interview schedules, and SLA warnings across all three user roles.

19 Admin & Operational Moderation Console

Provides platform operators with real-time vector matching health, job/company moderation queues, abuse reports, and SLA compliance monitoring.

20 Monetization Strategy (Current Proposed Pricing)

Revenue Model	Proposed Pricing	Scope & Commercial Terms
10% Placement Commission	10% First-Year CTC	Pre-authorized digital agreement. Invoiced upon verified candidate placement. (TBD: Payment terms & refund window).
Recruiter Pro Subscription	\$199 / recruiter / mo	Unlimited job postings, 25 monthly blind marketplace bids, custom SMTP connection, priority matching.
Enterprise ATS Bridge	\$999+ / company / mo	Greenhouse/Lever sync, dedicated account manager, custom SLAs, multi-seat team management.

21 Technical Architecture Model

React 19 (Vite) → Cloudflare Pages → Cloudflare Workers / Functions → AI APIs / Business Logic / Email Service → Supabase (PostgreSQL, Auth, Storage, Realtime).
Note: `server.ts` serves as a local development proxy and test harness.

22 Supabase Database Schema (13 Core Tables)

Table Name	Key Columns & Constraints	Security / RLS Rule
profiles	id (PK), email (UNIQUE), role (CHECK), created_at, updated_at	Self-read/write; role immutable by client
candidates	id (PK), user_id (FK), skills[], commission_agreement_signed, signature_hash	Candidate owner write; public read (non-PII)
companies	id (PK), user_id (FK), company_name, email_integration (JSONB), is_verified	Company owner write; public browsable
jobs	id (PK), company_id (FK), title, salary, required_skills[], status (CHECK)	Recruiter write; public read (active jobs)
applications	id (PK), job_id (FK), candidate_id (FK), status, match_score, timeline	Candidate (own) & Company (own jobs) only
blind_talent_profiles	id (PK), candidate_id (FK, UNIQUE), anonymous_handle, verified_skills	Public read (PII strictly scrubbed)
talent_bids	id (PK), blind_talent_id (FK), company_id (FK), salary_offer, expires_at	Target candidate & bidding company only
swipe_interactions	id (PK), candidate_id (FK), job_id (FK), action (CHECK), created_at	Candidate owner & system only
company_slas	company_id (FK, UNIQUE), avg_response_hours, badge_tier, ghosting_rate	Public read; system computed
notifications	id (PK), recipient_id, role, title, message, is_read	Recipient owner only
email_templates	id (PK), company_id (FK), category, subject, body_template	Company owner only
whatsapp_templates	id (PK), company_id (FK), category, message_template	Company owner only
admin_reports	id (PK), target_type, target_id, reason, status	Reporter insert; admin review only

23 Security, Data Privacy & Governance

- PII Isolation:** Candidate names, emails, and phones are excluded from all blind marketplace API responses.
- Credential Protection:** SMTP passwords and API secrets stored securely; never returned in client bundles.
- Data Governance:** Candidate profile export and deletion mechanisms compliant with applicable data privacy regulations.

24 API Security & Validation Standards

All protected endpoints must enforce Authentication, Authorization, Input Validation, Rate Limiting, and Structured Error Handling. Stack traces and database errors are suppressed from client responses.

25 AI Reliability & Provider Fallbacks

- Payload Limits:** Maximum upload size of 25MB on resume parser endpoints.
- Timeout & Retries:** 8-second AI timeout with exponential backoff and schema fallback defaults.

26 Background Processing Architecture

Heavy operations (bulk resume parsing, batch matching, email campaigns, report generation) execute asynchronously through queue/worker architectures to preserve sub-second UI responsiveness.

27 Audit Logging & Observability

Immutable audit logs record security events, application status changes, offer creations, marketplace reveals, and agreement signings without capturing raw credentials.

28 Performance & Scalability SLAs

- Card Swipe Response:** ≤ 16ms render time (60 FPS smooth motion).
- Algorithmic Match Scoring:** ≤ 50ms computation latency.
- Resume Parsing & Structuring:** ≤ 5.0s turnaround time.

29 Testable Acceptance Criteria (AC)

AC-AUTH — Authentication & Role Isolation Given an authenticated candidate user, when they attempt to access recruiter routes or API endpoints, then the server must reject the request with HTTP 403 and the UI must redirect to the candidate radar.
AC-AGREE — Mandatory Placement Agreement Given a candidate who has not signed the agreement, when they attempt to browse radar jobs or list a blind profile, then access is blocked until a valid SHA-256 e-signature record is verified server-side.
AC-MARKET-01 — Blind Marketplace Anonymity Given a recruiter browsing the blind marketplace, when they inspect a candidate profile, then the API response must not contain candidate name, email, phone, or photo.
AC-MARKET-02 — 72-Hour Bid Expiration Given a marketplace bid past its <code>expires_at</code> timestamp, when a candidate attempts to accept it, then the server must reject the action with an <code>EXPIRED_BID</code> error.
AC-MARKET-03 — Authorized Identity Reveal Given a valid, unexpired bid, when the candidate clicks "Accept", then candidate contact information is unmasked to the bidding company and an application record is initialized.
AC-MATCH — Algorithmic Match Calculation Given a candidate and a job opening, when match scoring runs, then the score must equal the normalized sum of domain, skill, role, location, experience, and salary weights.
AC-ATS — Kanban Drag-and-Drop Transition Given an applicant card dragged to a new stage, when dropped, then the application status updates in PostgreSQL, an audit record is created, and the candidate is notified.
AC-SLA — Constructive Feedback on Rejection Given an applicant rejected by a recruiter, when the rejection is confirmed, then the system generates structured gap feedback and a 30-day learning roadmap for the candidate.

30 Non-Functional Requirements (NFRs)

- Security:** Full RLS coverage, parameterized database queries, strict CORS, and encrypted credential storage.
- Accessibility:** WCAG 2.1 AA compliance, full keyboard navigation, accessible form inputs, and sufficient contrast ratios.
- Observability:** Structured JSON application logs, error telemetry, and vector latency metrics.

31 Current Release Scope (Phase 1 — Production)

In Scope (Current Release) - AI Resume OCR Parser (Affinda / Eden AI). 10% Placement E-Signature Engine. - Career Radar Swipe Deck & Matching Engine. 72-Hour Blind Reverse Marketplace. - Recruiter ATS & Kanban Pipeline. - Candidate Comparison Arena & AI Verdict. - Custom SMTP Connection Testing & WhatsApp. - Anti-Ghosting SLA & Constructive Feedback.	Out of Scope for Current Release - Autonomous AI voice screening calls. - Third-party ATS bidirectional sync (Greenhouse, Lever). - Web3 cryptographic skill credentials. - Multi-currency global salary bidding. - Native iOS and Android mobile binaries.
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32 Future Roadmap

- Phase 2 (Q3 2026):** Automated AI Voice Interview Screener, Webhook alerts (Slack/Discord), Greenhouse/Lever ATS connectors.
- Phase 3 (Q4 2026):** Cryptographic Proof-of-Work skill credentials, multi-currency global compensation, and native mobile apps.

33 Risk Matrix & Mitigation Strategies

Identified Risk	Severity	Engineering & Operational Mitigation
Blind Identity Leak	CRITICAL	Zero-PII server-side sanitization. Database RLS prevents unauthorized queries before bid acceptance.
Recruiter Ghosting	HIGH	Public SLA trust badges (Gold/Silver/Bronze) and automated stage escalation triggers.
Commission Dispute	MEDIUM	Tamper-evident signing record with SHA-256 hash and IP stamp. Pre-production legal review.

34 Open Product & Legal Decisions

Decision Item	Owner	Status	Impact & Considerations
Commission Payer Model	Product / Legal	TBD	Determine whether 10% fee is candidate-authorized direct fee or employer-invoiced placement fee.
Refund & Guarantee Window	Finance / Legal	TBD	Define 30/60/90 day candidate retention and replacement policy upon early employment termination.
Personalization Threshold	Data / Engineering	IN REVIEW	Establish minimum swipe interaction count (e.g. 15 swipes) before personalized re-ranking activates.

35 Related Engineering Documents

This PRD is supported by companion technical specifications in the project repository: `ARCHITECTURE.md`, `DATABASE.md`, `SECURITY.md`, `API_SPEC.md`, `ACCEPTANCE_CRITERIA.md`, and `ROADMAP.md`.

36 Internal Sign-Off & Verification

PRODUCT ARCHITECTURE SwipeHired Product Lead Status: Engineering Ready · Aug 2026	FULL-STACK ENGINEERING SwipeHired Engineering Lead Status: Implementation Verified · Aug 2026	SECURITY & GOVERNANCE Platform Security Lead Status: Security Boundaries Approved · Aug 2026
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